C27 Aromatic Chemistry – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Benzene **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Molecular formula of benzene				C_6H_6
2	Bond angle in benzene				120°
3	C–C bond length in benzene				Between C–C and C=C
4	Bonding in benzene				Ring of delocalised π electrons
5	Enthalpy of hydrogenation of benzene vs Kekulé				Less exothermic than expected
6	Expected enthalpy of hydrogenation of cyclohexa-1,3,5-triene				$-360\ kJ\ mol^{-1}$
7	Why benzene undergoes substitution not addition				Keeps stable delocalised ring
8	Mechanism type for benzene				Electrophilic substitution
9	Name of the C_6H_5– group				Phenyl
10	Why benzene is resistant to addition				Delocalisation energy (stability)

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Nitration and Acylation **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Reagents for nitration				Conc HNO_3 + conc H_2SO_4
2	Temperature for mononitration				Below 55 °C
3	Electrophile in nitration				NO_2^+
4	Equation forming NO_2^+				$HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$
5	Uses of nitration				Explosives; making amines for dyes
6	Reagents for Friedel–Crafts acylation				Acyl chloride + $AlCl_3$
7	Electrophile in acylation				RCO^+
8	Equation forming the acylium ion				$RCOCl + AlCl_3 \rightarrow RCO^+ + AlCl_4^-$
9	Equation regenerating the catalyst				$H^+ + AlCl_4^- \rightarrow AlCl_3 + HCl$
10	Product of benzene + ethanoyl chloride				Phenylethanone

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